

Relation Algebra

§1. Notation

Traditionally Adjunction $F \dashv G$ is defined as $F(x) \leq y$ iff $x \leq G(y)$, and you derive laws like this:

$$\begin{aligned} F(x) \leq F(x) &\implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x))) \\ G(y) \leq G(y) &\implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y) \end{aligned}$$

By define $X : * \mapsto x$, $Y : * \mapsto y$ and use diagram order, application can be replaced by composition: $XF \leq Y$ iff $X \leq YG$

(1a)

¹ • write F as $/$, G as $\backslash$

² • render $a \leq b$ as $\boxed{\frac{a}{b}}$

³ • **rewrite** $X \leq XF$ as $1 \leq F$

$$\begin{aligned} \boxed{\frac{X/}{Y}} &= \boxed{\frac{X}{Y\backslash}} && \text{adj} \\ \\ \boxed{\frac{X/}{X/}} &\xRightarrow{\text{adj}} \boxed{\frac{X}{X\wedge}} \xRightarrow{\text{rewrite}} \boxed{\frac{*}{\wedge}}_{\text{unit}} \xRightarrow{\text{put } \backslash \text{ on left}} \boxed{\frac{\backslash}{\wedge\backslash}} \\ &\xRightarrow{\text{put } / \text{ on right}} \boxed{\frac{/}{\wedge/}} \\ \\ \boxed{\frac{Y\backslash}{Y\backslash}} &\xRightarrow{\text{adj}} \boxed{\frac{Y\vee}{Y}} \xRightarrow{\text{rewrite}} \boxed{\frac{\vee}{*}}_{\text{counit}} \xRightarrow{\text{put } / \text{ on left}} \boxed{\frac{\vee/}{/}} \\ &\xRightarrow{\text{put } \backslash \text{ on right}} \boxed{\frac{\vee\backslash}{\backslash}} \end{aligned}$$

now we have $\wedge/ = /$ and $\vee\backslash = \backslash$, and you just discovered string diagrams.

§1.1. Adjunctions

(1.1a)

	F $\dashv$ G	F 's type	monad FG	1 $\sqsubseteq$ FG	GF $\sqsubseteq$ 1	F preserves u	G preserves n	FGF = F	GFG = G
1	$\triangleleft \dashv \triangleright$	$A \rightarrow A \otimes A$	$\triangleleft \triangleright = 1$	$1 \sqsubseteq \triangleleft \triangleright$	$\triangleright \triangleleft \sqsubseteq 1$	$(R \cup S) \triangleleft = R \triangleleft \cup S \triangleleft$	$(R \cap S) \triangleright = R \triangleright \cap S \triangleright$	$\triangleleft \triangleright \triangleleft = \triangleleft$	$\triangleright \triangleleft \triangleright = \triangleright$
2	$\multimap \dashv \multimap$	$A \rightarrow I$	$\multimap \multimap = T$	$1 \sqsubseteq \multimap \multimap$	$\multimap \multimap \sqsubseteq 1$	$(R \cup S) \multimap = R \multimap \cup S \multimap$	$(R \cap S) \multimap = R \multimap \cap S \multimap$	$\multimap \multimap \multimap = \multimap$	$\multimap \multimap \multimap = \multimap$
3	$\circ \dashv \circ$	$(A \rightarrow B) \rightarrow (B \rightarrow A)$	$R^{\circ \circ} = R$	$R^{\circ \circ} = R$	$R^{\circ \circ} = R$	$(R \cup S)^{\circ} = S^{\circ} \cup R^{\circ}$	$(R \cap S)^{\circ} = R^{\circ} \cap S^{\circ}$	$R^{\circ \circ} = R$	$R^{\circ \circ} = R$
4	$\circ \triangleleft \dashv \triangleright \circ$	$(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$	1	$(\circ \triangleleft \otimes 1) (1 \otimes \triangleright \circ) = 1$	$(1 \otimes \circ \triangleleft) (\triangleright \circ \otimes 1) = 1$	$=$	$=$	$=$	$=$
5	$\triangleright \circ \dashv \circ \triangleleft$	$(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$	1	$(1 \otimes \circ \triangleleft) (\triangleright \circ \otimes 1) = 1$	$(\circ \triangleleft \otimes 1) (1 \otimes \triangleright \circ) = 1$	$=$	$=$	$=$	$=$
6	$\Delta \dashv \cap$	$(A \rightarrow B) \rightarrow (A \rightarrow B)^2$	$R \cap R = R$	$R \cap R = R$	$R \cap S \sqsubseteq R$	$\Delta(R \cup S) = \Delta R \cup \Delta S$	$(R \cap T) \cap (S \cap U) = (R \cap S) \cap (T \cap U)$	$R \cap R = R$	$R \cap R = R$
7	$\cup \dashv \Delta$	$(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$	$(R, S) \mapsto (R \cup S, R \cup S)$	$R \sqsubseteq R \cup S$	$R \cup R = R$	$(R \cup T) \cup (S \cup U) = (R \cup S) \cup (T \cup U)$	$\Delta(R \cap S) = \Delta R \cap \Delta S$	$R \cup R = R$	$R \cup R = R$
8	$\emptyset \dashv !$	$\{*\} \rightarrow (A \rightarrow B)$			$\emptyset \sqsubseteq R$				
9	$\cdot S \dashv / S$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	S/S	$R \sqsubseteq (RS)/S$	$(T/S)S \sqsubseteq T$	$(S \cup T)R = (SR) \cup (TR)$	$(R \cap R') / S = (R / S) \cap (R' / S)$	$((RS)/S)S = RS$	$((T/S)S)/S = T/S$
10	$S \cdot \dashv S \setminus$	$(B \rightarrow C) \rightarrow (A \rightarrow C)$	$S \setminus S$	$R \sqsubseteq S \setminus (SR)$	$S(S \setminus T) \sqsubseteq T$	$R(S \cup T) = (RS) \cup (RT)$	$S \setminus (R \cap R') = (S \setminus R) \cap (S \setminus R')$	$S(S \setminus (SR)) = SR$	$S \setminus (S(S \setminus T)) = S \setminus T$
11	$R \cap \dashv R \Rightarrow$	$(A \rightarrow B) \rightarrow (A \rightarrow B)$	$X \mapsto R \Rightarrow (X \cap R)$	$X \sqsubseteq R \Rightarrow (X \cap R)$	$R \cap (R \Rightarrow Y) \sqsubseteq Y$	$R \cap (X \cup Y) = (R \cap X) \cup (R \cap Y)$	$R \Rightarrow S \cap T = (R \Rightarrow S) \cap (R \Rightarrow T)$	$R \cap (R \Rightarrow (X \cap R)) = X \cap R$	$R \Rightarrow (R \cap (R \Rightarrow Y)) = R \Rightarrow Y$
12	$\mathcal{D} \dashv \cdot T$	$(A \rightarrow B) \rightarrow \text{Cor } A$	$R \mapsto (\mathcal{D}R)T$	$R \sqsubseteq (\mathcal{D}R)T$	$\mathcal{D}(AT) \sqsubseteq A$	$\mathcal{D}(R \cup S) = \mathcal{D}R \cup \mathcal{D}S$	$(A \cap B)T = AT \cap BT$	$\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$	$(\mathcal{D}(AT))T = AT$
13	$\mathcal{R} \dashv T \cdot$	$(A \rightarrow B) \rightarrow \text{Cor } B$	$R \mapsto T(\mathcal{R}R)$	$R \sqsubseteq T(\mathcal{R}R)$	$\mathcal{R}(TA) \sqsubseteq A$	$\mathcal{R}(R \cup S) = \mathcal{R}R \cup \mathcal{R}S$	$T(A \cap B) = TA \cap TB$	$\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$	$T(\mathcal{R}(TA)) = TA$
14	$\cdot f \dashv \cdot f^{\circ}$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	ff°	$\text{Map}(f) \Rightarrow 1 \sqsubseteq f f^{\circ}$	$f^{\circ} f \sqsubseteq 1$	$(S \cup T)R = (SR) \cup (TR)$	$\text{Simple}(g) \Rightarrow (X \cap Y)g^{\circ} = (Xg^{\circ}) \cap (Yg^{\circ})$	$ff^{\circ}f = f$	$f^{\circ}ff^{\circ} = f^{\circ}$
15	$f^{\circ} \cdot \dashv f \cdot$	$(A \rightarrow C) \rightarrow (B \rightarrow C)$	ff°	$\text{Map}(f) \Rightarrow 1 \sqsubseteq f f^{\circ}$	$f^{\circ} f \sqsubseteq 1$	$R(S \cup T) = (RS) \cup (RT)$	$f(X \cap Y) = fX \cap fY$	$ff^{\circ}f = f$	$f^{\circ}ff^{\circ} = f^{\circ}$
16	$i \dashv E$	$\text{Map} \multimap \text{Rel}$	E	$\frac{1}{\exists} : A \rightarrow PA$	$\exists : PB \rightarrow B$	—	—	$\frac{\exists}{\exists} = 1$	$\frac{R}{\exists} \exists = R$
17	$\cdot \exists \dashv \frac{\exists}{\exists}$	$\text{Map}(A, PB) \rightarrow (A \rightarrow B)$	1	$\frac{f \exists}{\exists} = f$	$\frac{R}{\exists} \exists = R$	—	—	$\frac{f \exists}{\exists} = f \exists$	$\frac{R}{\exists} \exists = R$
18	$\frac{\exists}{\exists} \dashv \cdot \exists$	$(A \rightarrow B) \rightarrow \text{Map}(A, PB)$	1	$\frac{R}{\exists} \exists = R$	$\frac{f \exists}{\exists} = f$	—	—	$\frac{\exists}{\exists} = \frac{R}{\exists}$	$\frac{f \exists}{\exists} = f \exists$

§1.2. Composing adjunctions

(1.2a)

$$\begin{array}{c}
\boxed{\begin{array}{c} X \\ f(R/T) \end{array}} \xleftrightarrow{f^\circ \cdot \dashv f} \boxed{\begin{array}{c} f^\circ X \\ R/T \end{array}} \xleftrightarrow{\cdot T \dashv / T} \boxed{\begin{array}{c} (f^\circ X)T \\ R \end{array}} \\
\text{f a map} \\
\xleftrightarrow{\text{associativity}} \boxed{\begin{array}{c} f^\circ(XT) \\ R \end{array}} \xleftrightarrow{f^\circ \cdot \dashv f} \boxed{\begin{array}{c} XT \\ fR \end{array}} \xleftrightarrow{\cdot T \dashv / T} \boxed{\begin{array}{c} X \\ (fR)/T \end{array}} \\
\xrightarrow{\text{indirect equality}} \boxed{\begin{array}{c} f(R/T) \\ (fR)/T \end{array}}
\end{array}$$

$$\text{Map}(f) \Rightarrow f(R/S) = (fR)/S$$

(1.2b)

	$\bullet T$	$T \bullet$	$\bullet g$	$g^\circ \bullet$	Tn	$^\circ$	$\text{key} \triangleleft$	Δ	
1	$\bullet S$	$R / (S_1 S_2)$ $= R / S_2 / S_1$	$S \setminus (R / W)$ $= (S \setminus R) / W$	$R / (Sg) =$ $(Rg^\circ) / S$	$\text{Map}(f) \implies f$ $(R / S) =$ $(f R) / S$	$---$	$(Y/S)^\circ =$ $S^\circ \setminus (Y^\circ)$	$(RS)^\wedge =$ $R^\wedge (S \otimes 1)$	$(R \cap R') /$ $S = (R / S)$ $\cap (R' / S)$
2	$S \bullet$	$S \setminus (R / W)$ $= (S \setminus R) / W$	$(S_1 S_2) \setminus R$ $= S_2 \setminus (S_1 \setminus R)$	$S \setminus (Rg^\circ) =$ $(S \setminus R)g^\circ$	$(g^\circ S) \setminus R =$ $S \setminus (gR)$	$---$	$(S \setminus Y)^\circ =$ Y° / S°	$((S \otimes 1)R)^\wedge =$ $SR^\wedge$	$S \setminus (R \cap$ $R') = (S \setminus$ $R) \cap (S \setminus$ $R')$
3	$\bullet f$	$\text{Map}(f) \implies$ $R / (f S) =$ $(R / S) f^\circ$	$T \setminus (Rf^\circ) =$ $(T \setminus R) f^\circ$	$(R S)^\circ = S^\circ$ R°			$(R S)^\circ = S^\circ$ R°		$\text{Simple}(g)$ $\implies (X \cap Y)$ $g^\circ = (X g^\circ)$ $\cap (Y g^\circ)$
4	$f^\circ \bullet$	$\text{Map}(f) \implies f$ $(R / S) =$ $(f R) / S$	$(Tf^\circ) \setminus R =$ $f(T \setminus R)$	$---$	$(R S)^\circ = S^\circ$ R°		$(R S)^\circ = S^\circ$ R°		$f(T_1 \cap T_2) =$ $fT_1 \cap fT_2$
5	Rn	$---$	$---$	$---$	$R \Rightarrow S \Rightarrow T =$ $R \cap S \Rightarrow T$	$(R \Rightarrow Y)^\circ =$ $R^\circ \Rightarrow (Y^\circ)$	$---$		$R \Rightarrow S \cap T =$ $(R \Rightarrow S) \cap$ $(R \Rightarrow T)$
6	$^\circ$	$(Y/T)^\circ =$ $T^\circ \setminus (Y^\circ)$	$(T \setminus Y)^\circ =$ Y° / T°	$(R S)^\circ = S^\circ$ R°	$(R S)^\circ = S^\circ$ R°	$(T \Rightarrow Y)^\circ =$ $T^\circ \Rightarrow (Y^\circ)$	$R^\circ \circ = R$	$(R^\wedge)^\circ =$ $(R^\circ \otimes 1)$ $(1 \otimes \triangleright \dashv)$ both bends: the ^\circ section's display	$(T_1 \cap T_2)^\circ =$ $T_1^\circ \cap T_2^\circ$
7	$\text{key} \triangleleft$	$(RT)^\wedge =$ $R^\wedge (T \otimes 1)$	$((T \otimes 1)R)^\wedge =$ $TR^\wedge$	$---$	$---$	$---$	$(R^\wedge)^\circ =$ $(R^\circ \otimes 1)$ $(1 \otimes \triangleright \dashv)$ both bends: the ^\circ section's display	$---$	$---$
8	u	$(S \cup T) R =$ $(S R) \cup (T R)$	$R (S \cup T) =$ $(R S) \cup (R T)$	$(S \cup T) R =$ $(S R) \cup (T R)$	$R (S \cup T) =$ $(R S) \cup (R T)$	$R \cap (S \cup T)$ $= R \cap S \cup R$ $\cap T$	$(R \cup S)^\circ =$ $S^\circ \cup R^\circ$		
9	$\emptyset$	$\emptyset R = \emptyset$	$R \emptyset = \emptyset$	$\emptyset R = \emptyset$	$R \emptyset = \emptyset$	$R \cap \emptyset = \emptyset$	$\emptyset^\circ = \emptyset$		

§1.3. Adjoint triples

Beyond $\mathbf{u} \dashv \mathbf{n}$, which §1.1 already carries as two rows, the table holds one adjoint triple with content: $\exists_f \dashv f^* \dashv \forall_f$ along a map f , once on each side of the composite. $f : A \rightarrow B$ throughout this subsection.

(1.3a)

$$\begin{aligned} & \cdot f \dashv \cdot f^\circ \dashv /f^\circ \\ & f^\circ \cdot \dashv f \cdot \dashv f \backslash \end{aligned}$$

The first two links of each chain are the map shunting rules, and the third is §1.1's division row read at a chosen divisor: $\cdot S \dashv /S$ at $S := f^\circ$, and $S \cdot \dashv S \backslash$ at $S := f$.

The third link is not the first over again. For a map f , $/f$ collapses to $\cdot f^\circ$; being a map is exactly what that collapse spends $\text{Map}(f) \implies R / (f \ S) = (R / S) f^\circ$ — f° is not a map, so $/f^\circ$ does not collapse in turn, and the three operators stay distinct. On $\text{Rel}(\mathbf{I}, A) = \mathbf{P}A$ they are the image triple:

(1.3b)

	operator	acts by	name
1	$\cdot f$	$S \mapsto f[S]$	direct image, $\exists_f$
2	$\cdot f^\circ$	$T \mapsto f^{-1}[T]$	inverse image, f^*
3	$/f^\circ$	$S \mapsto \{b : f^{-1}(b) \subseteq S\}$	$\forall_f$

§1.1's $\mathcal{D} \dashv \cdot \tau$ is this same chain along the projection $A \otimes B \rightarrow A$, read through $\text{Rel}(A, B) = \mathbf{P}(A \otimes B)$: $\mathcal{D}R = \{(a, a) : \exists b. aRb\}$, $\mathbf{A}T$ is the relation that ignores b altogether, and the third link is $R \mapsto \mathbf{1} \cap R / \tau = \{(a, a) : \forall b. aRb\}$.

(1.3c)

$$\mathcal{D} \dashv \cdot \tau \dashv \mathbf{1} \cap \cdot / \tau$$

Three is where it stops, and one map breaks both ends. Take $A = \{a_1, a_2\}$ and $B = \{b\}$: $f[\{a_1\}] \cap f[\{a_2\}] = \{b\}$ while $f[\{a_1\} \cap \{a_2\}] = \emptyset$, so $\cdot f$ does not preserve meets and has no left adjoint, and the same two subsets give $\forall_f(\{a_1\} \cup \{a_2\}) = \{b\}$ against $\forall_f\{a_1\} \cup \forall_f\{a_2\} = \emptyset$, so $/f^\circ$ does not preserve joins and has no right adjoint. A monic f restores the binary case and no more — τf still reaches only $\text{im } f$, and $\forall_f \emptyset = B \setminus \text{im } f$ is still not $\emptyset$. The chain extends only when f° is a map as well, and then $/f^\circ = \cdot f^\circ = \cdot f$ and it repeats forever. For an S with neither S nor S° a map there is no triple at all: $\cdot S \dashv /S$ is two links and stops.

The rows that chain forever say nothing by it: $^\circ$ is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and $^\circ \dashv \cdot \dashv \cdot \dashv \cdot$ has no end, and §1.1's other row of that kind, $\circ \dashv \triangleleft \dashv \triangleright \dashv \circ$, goes the same way.

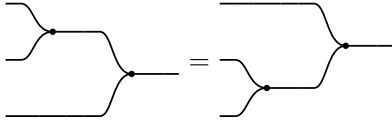
§2. Relations

(2a)

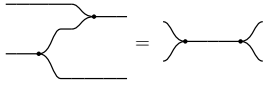
$\mathbf{Rel}$ is a poset-enriched category with $(\otimes, \circ)$ where $\otimes$ is commutative cartesian product, and converse $\circ^{-1}$, and

$\mathbf{C}: (\triangleright : A \otimes A \rightarrow A, \circ : I \rightarrow A)$ a commutative monoid with $\mathbf{C}^\circ$. We write $\mathbf{C}^\circ$ as $(\triangleleft : A \rightarrow A \otimes A, \circ : A \rightarrow I)$

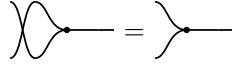
(2b)



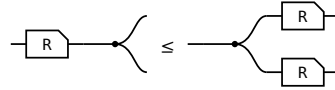
$\triangleright$ associative



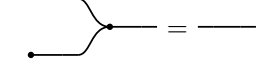
Frobenius, one half — the other is its $\circ$



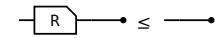
$\triangleright$ commutative



$R \triangleleft \leq \triangleleft (R \otimes R)$



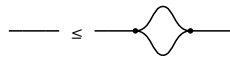
$\circ$ is its unit



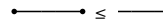
$R \circ \leq \circ$

§2.1. $\forall$ object A , $(A, \triangleleft, \multimap) \dashv (A, \triangleright, \multimap)$ 

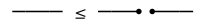
$$\triangleright \triangleleft \leq 1$$



$$1 \leq \triangleleft \triangleright$$



$$\bullet - \multimap \leq 1$$



$$1 \leq \multimap \bullet$$

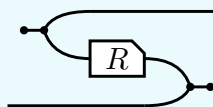
(2.1a)

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In **Rel** it reads $\{(a, a)\} \subseteq A \times A$ — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

§3. $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

(3a)

$\circ$ is primitive, part of the data the first section lists. It turns both of R 's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are R° , not its definition:



$$R^\circ = (\circ \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \multimap \circ)$$

where $\circ \triangleleft : \mathbb{I} \rightarrow a \otimes a$ opens a pair of wires out of nothing and $\triangleright \circ : b \otimes b \rightarrow \mathbb{I}$ closes one, so the input of R° is where the output of R was. Taking that formula as the definition would now be circular: $\triangleleft$ and $\multimap \circ$ are $\triangleright^\circ$ and $\circ \triangleleft$. The laws of $\circ$ are that it is a contravariant 2-functor $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$:

$$(i) \ 1^\circ = 1 \quad (ii) \ (RS)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§3.1. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:

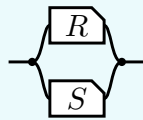
$$\begin{array}{c}
 \text{---} \quad \text{---} \quad \text{---} \\
 \quad \quad \quad \diagup \quad \diagdown \\
 \quad \quad \quad \bullet \quad \bullet \\
 \quad \quad \quad \diagdown \quad \diagup \\
 \quad \quad \quad \bullet \quad \bullet \\
 \quad \quad \quad \text{---} \quad \text{---} \quad \text{---}
 \end{array}
 =
 \begin{array}{c}
 \text{---} \quad \text{---} \quad \text{---} \\
 \quad \quad \quad \diagup \quad \diagdown \\
 \quad \quad \quad \bullet \quad \bullet \\
 \quad \quad \quad \diagdown \quad \diagup \\
 \quad \quad \quad \bullet \quad \bullet \\
 \quad \quad \quad \text{---} \quad \text{---} \quad \text{---}
 \end{array}
 \quad (3.1a)$$

R° is DEFINED as the bending of $(R \otimes 1) \triangleright \bullet$, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the $\bullet \leftarrow \triangleright$ bends the **a** strand down and the $\triangleright \bullet$ brings it back up, while the **b** strand rides through untouched. Nothing else is spent below.

§4. $\cap$ is a commutative idempotent monoid on every hom-set

(4a)

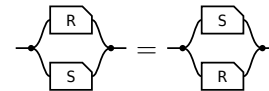
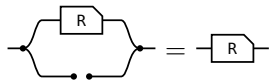
The **meet** of $R, S : a \rightarrow b$, the paper's **convolution**, is $R \cap S := \langle \downarrow (R \otimes S) \rangle$ — copy the input, run R and S on the two copies, merge the results — so what comes out is what both of them do.



transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent $\cap$, with unit the maximal arrow $T = \text{---} \circ \text{---}$ (the paper's Lemma 4.11).

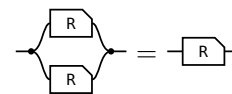
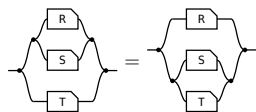
(4b)



unit: one half of T per end — the merge's unit law absorbs the $\circ$, the copy's

commutative: σ crosses $R \otimes S$ by naturality and is absorbed by cocommutativity and commutativity

counit law the $\circ$



associative: coassociativity and associativity; $\otimes$ re-brackets for nothing, **idempotent:** the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So $\leq$ is the order this monoid induces. $R \cap S \leq R$ comes from the unit, and idempotency turns anything under both S and T into something under $S \cap T$, since $R = R \cap R \leq S \cap T$.

And one law relating $\cap$ to composition, which is **not** an equation:

(4c)

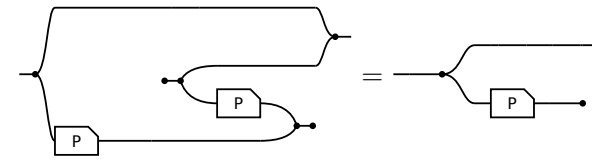
semi-distributivity, and what supplies it	picture
$R \cap (S \cap T) \leq R \cap S \cap T$ — the lax copy law. Equality exactly when R is single valued: the Maps section's $F(R \cap S) = F R \cap F S$.	

§5. Domain and range

(5a)

The **domain** $\text{dom}(R) \triangleq \mathbb{1} \sqcap (R \circ R^\circ)$ and the **range** $\text{ran}(R) \triangleq \text{dom}(R^\circ)$.

(5b)



$\triangleright$ lands the return leg back on the value $\triangleleft$ handed out,
so $R^\circ \triangleright$ cuts to $\circlearrowleft$ — Frobenius

Running R and throwing the result away leaves only the fact that R could fire, and $\text{ran}(R)$ is the same picture with the box mirrored. In Rel both steps are $\{(a, a) : \exists b. a R b\}$.

(5c)

1	$\text{dom}(R) \triangleq \mathbb{1} \sqcap (R \circ R^\circ)$
2	$\text{Coreflexive}(X) \implies \text{dom}(R) \sqsubseteq X \iff R \sqsubseteq X \circ R$
3	$\text{dom}(R \circ S) \sqsubseteq \text{dom}(R)$
4	$\text{dom}(R \sqcap S) = \mathbb{1} \sqcap (S \circ R^\circ)$
5	$R \text{ entire} \iff \text{Entire}(R) \triangleq \text{dom}(R) = \mathbb{1}$
6	$R \text{ simple} \iff \text{Simple}(R) \triangleq R^\circ \circ R \sqsubseteq \mathbb{1}$
7	$R \text{ a map} \iff \text{Map}(R) \triangleq \text{Entire}(R) \wedge \text{Simple}(R)$
8	$\text{Entire}(R) \wedge \text{Entire}(S) \implies \text{Entire}(R \circ S)$ $\text{Simple}(f) \wedge \text{Simple}(g) \implies \text{Simple}(f \circ g)$ $\text{Map}(f) \wedge \text{Map}(g) \implies \text{Map}(f \circ g)$
9	$\text{Entire}(R \circ S) \implies \text{Entire}(R)$

§5.1. Sliding the discard

$\text{dom}(RS) \sqsubseteq \text{dom}(R)$, and a single glyph for **dom** would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

$$\begin{array}{c}
 \text{---} \begin{array}{c} \nearrow \\ \searrow \end{array} \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \boxed{R} \\ \boxed{S} \end{array} \text{---} \bullet \quad \leq \quad \text{---} \begin{array}{c} \nearrow \\ \searrow \end{array} \begin{array}{c} \text{---} \\ \text{---} \end{array} \boxed{R} \text{---} \bullet \\
 S \multimap \sqsubseteq \multimap, \text{ the lax axiom for } \multimap \text{ in the first section} \\
 \text{--- the discard slides back past } S
 \end{array}
 \tag{5.1a}$$

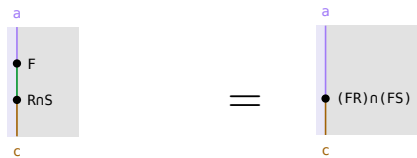
Equality is S entire, which is the same picture read as $\text{dom}(R) = \mathbb{1} \iff R$ entire.

§6. Maps

(6a)

1 surjective $1 \subseteq R^\circ R$, that is, R° entire.	single valued $R^\circ R \subseteq 1$
2 entire $1 \subseteq RR^\circ$. With single valued, a map.	injective $RR^\circ \subseteq 1$

(6b)

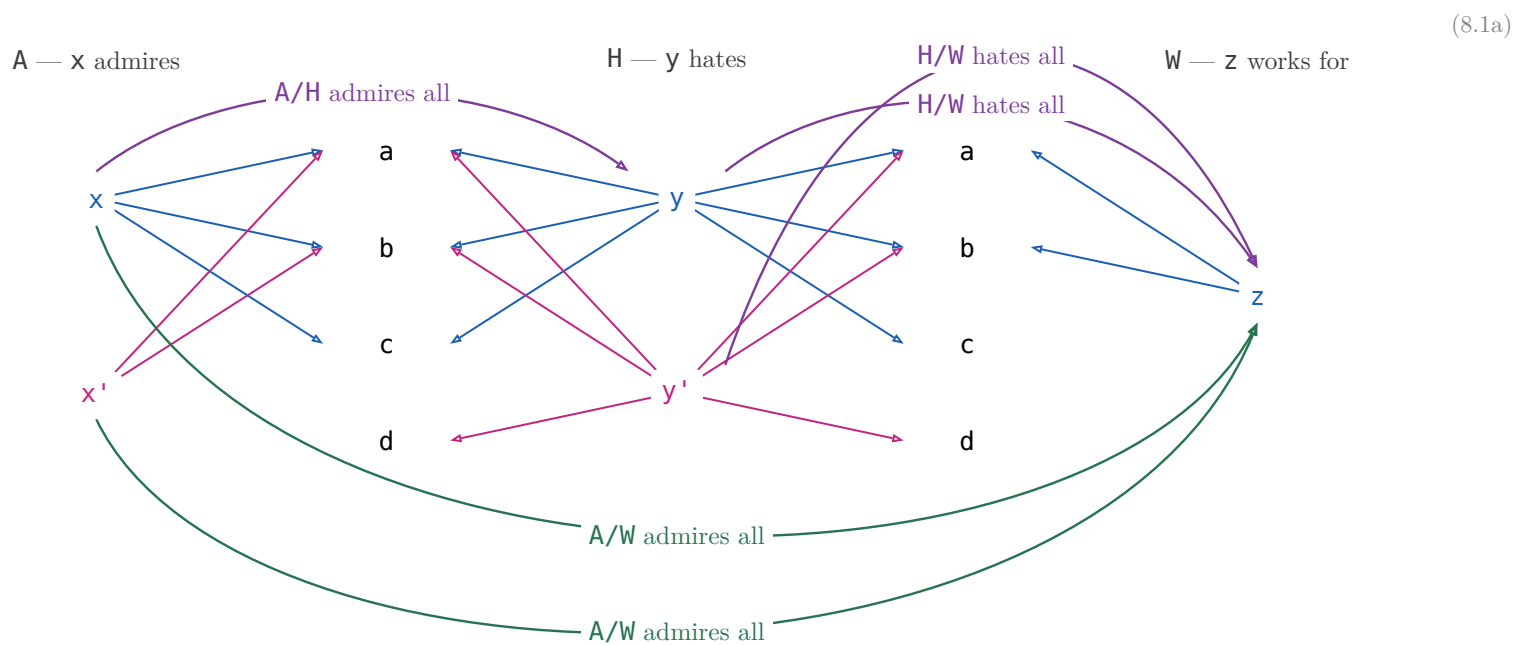
law	picture
$\text{Simple}(F) \implies F(R \cap S) = (F R) \cap (F S)$ F single valued	 one person who admires both a at A, b at B

§8. / is all of

(8a)

$x (R/S) y \iff \forall p. y S p \rightarrow x R p$
 $x (S \setminus R) y \iff \forall p. p S x \rightarrow p R y$

/compares images: $S(y) \subseteq R(x)$. \ compares preimages: $S^\circ(x) \subseteq R^\circ(y)$.
example: A admires, H hates, W works for
 $x (A/H) y$ – x admires everyone y hates.
 $x (H \setminus A) y$ – everyone who hates x admires y.

§8.1. $(R/S)(S/W) \sqsubseteq R/W$ 

(8.1b)

$x (A/H) y \rightarrow x \text{ admires everyone } y \text{ hates}$
 $y (H/W) z \rightarrow y \text{ hates everyone } z \text{ works for}$
 $x (A/W) z \rightarrow x \text{ admires everyone } z \text{ works for}$

$(A/H)(H/W)$ is a path: $x \rightarrow y \rightarrow z$, and that is all of it. A/W also holds of x' , who admires everyone z works for — but x' does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of $(R/S)(S/W) \sqsubseteq R/W$.

1	$X \sqsubseteq R/S \iff XS \sqsubseteq R$ X is any x -to- y pairing; one that only pairs x with a y such that x admires everyone y hates lies inside R/S , and R/S is the largest such.	
2	$X \sqsubseteq S \setminus R \iff SX \sqsubseteq R$ The mirror — divide on the left when x comes first.	
3	$(R/S)S \sqsubseteq R$ There is a y such that x admires everyone y hates, and p is one of the people y hates — then x admires p too. Strict at $S=\emptyset$: R/S is everyone, $(R/S)S=\emptyset$.	
4	$S (S \setminus R) \sqsubseteq R$ The mirror.	
5	associate: $R/(S_1 S_2) = (R/S_2)/S_1$ A friend's enemy is two hops: divide by the far end first.	
6	$(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$ The mirror.	
7	maps: $f (R/S) = (fR)/S$ Rename x before or after dividing — the licence to write fR/S .	
8	$R/(fS) = (R/S) f^\circ$ Rename y : a map leaves a denominator as f° outside the box.	
9	$(R/S) (S/W) \sqsubseteq R/W$ Someone who admires all of a hate-set that already covers everyone z works for admires those people too.	
10	$1 \sqsubseteq R/R$ R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.	
11	$(R/R) (R/R) = R/R$ R/R is the preorder admires at least as much as , and a preorder is idempotent. Freyd writes $\sqsubseteq$; with $1 \sqsubseteq R/R$ above it is an equality.	
12	$(R/R)R = R$ Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.	
13	$R/1 = R$ Dividing by 1 : p 's set is just $\{p\}$, so admiring all of it is admiring p .	
14	$R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$ Admiring a combined hate-set is admiring each set in full.	
15	$S \setminus (R/W) = (S \setminus R)/W$ Which is why $S \setminus R/W$ needs no bracket.	

Fifteen laws, fifteen pictures, and not one shows a generator: $\cap$, $\cup$, $\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.

§9. $\frac{R}{S}$

(9a)

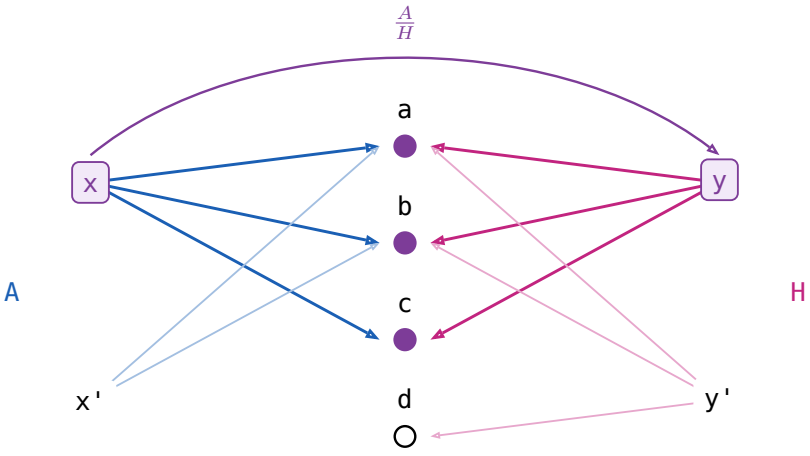
$R /_s S \triangleq (R / S) \cap (S / R)^\circ$ what R sends x to and nothing else is what S sends y to. In $\mathbf{Rel}$ x and y has the same image: $\forall p. (x R p \iff y S p)$

(9b)

$x (A/H) y$ — x admires everyone y hates
 $x ((H/A)^\circ) y$ — x admires only people y hates
 $x ((A/H) \cap (H/A)^\circ) y$ — x admires only and all whom y hates

x admires exactly whom y hates: $/$ is **all of**, $\frac{R}{S}$ is **only all of**.

(9c)



	law	the reading
1	$X \sqsubseteq R /_s S \iff X S \sqsubseteq R \wedge X^\circ R \sqsubseteq S$ X sits inside the matching exactly when it carries S back inside R and, reversed, carries R back inside S.	X may pair x with y only when x admires exactly whom y hates. Both halves must typecheck, so the operation is partial .
2	$(R /_s S)^\circ = S /_s R$ reversing the matching swaps the two relations.	Matching is symmetric.
3	$(R /_s S) (S /_s W) \sqsubseteq R /_s W$ a match from R to S followed by one from S to W is a match from R to W.	And transitive.
4	$(R /_s S) S \sqsubseteq R$ following the matching by S lands inside R.	$(\exists y. x(\frac{R}{S})y \wedge ySp) \rightarrow xRp$ x only admires whom y hates $\frac{R}{S}S = \text{Dom}(\frac{R}{S})R$
5	$(R /_s R) R = R$ matching R against itself and then following R gives R back.	$(\exists y. x(\frac{R}{R})y \wedge yRp) \iff xRp$ x and y admire the same people y=x always qualifies ($1 \sqsubseteq R\%R$ below)
6	$1 \sqsubseteq R /_s R$ every x matches itself.	$x(\frac{R}{R})y$ if x and y admires the same peoples.
7	$(R /_s R) (R /_s R) = R /_s R$ matching twice matches no more pairs than matching once.	So the relation admires the same people is an equivalence relation.
8	$\text{Symmetric}(X) \implies X \sqsubseteq R /_s R \iff X R \sqsubseteq R$ a symmetric X sits inside the matching exactly when following it by R adds nothing to R.	The largest symmetric arrow that leaves R alone.
9	$\text{simplePart}(R) \triangleq R /_s 1$ matching R against the identity keeps the X whose image is a single point.	The people who admire exactly one person and nobody else. It equals R only when R is simple, unlike $R/1=R$.
10	$\text{dom}(R /_s S) = 1 \cap ((R / S) (S / R))$ the X that match something are those the two one-sided divisions take round the loop back to X.	Its domain is the domain of simplicity of R.

§10. Freyd's Power allegories

(10a)

A **power allegory** is a division allegory with one unary operation on arrows, $\exists$ **epsilon**, subject to

$$\begin{array}{ll} \exists_R \square = R\square, & \exists_R = \exists_R \square \\ \mathbb{1} \sqsubseteq (R/\exists_R)(\exists_R/R) & \exists_R \text{ is } \mathbf{thick} \\ \exists_R = \mathbb{1} & \exists_R \text{ is } \mathbf{straight} \end{array}$$

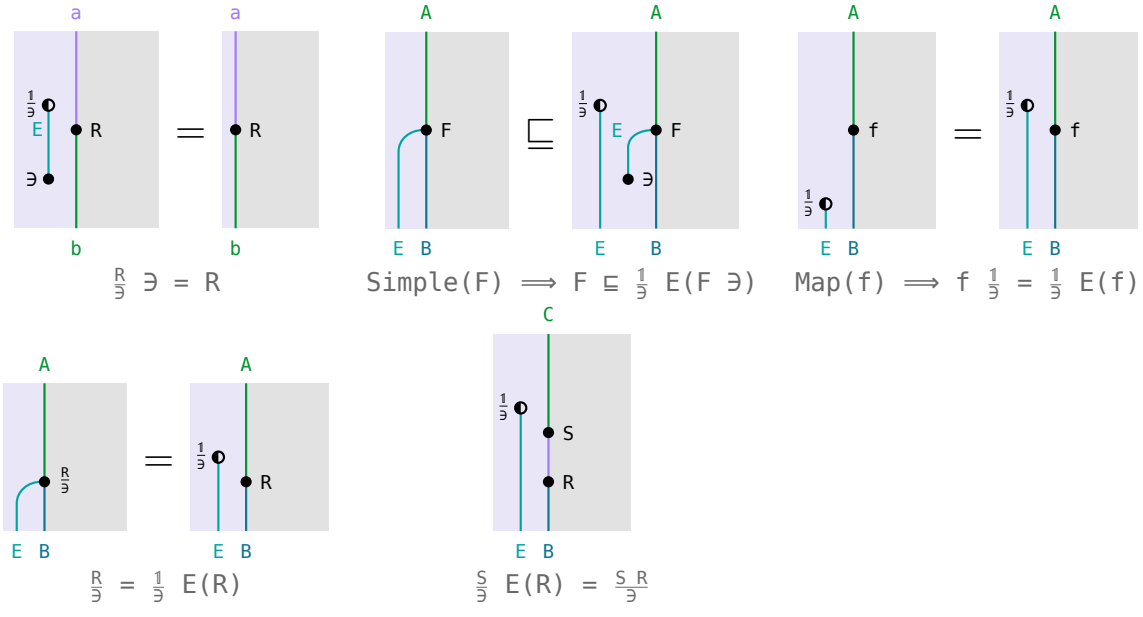
$R\square$ is R 's target, an identity arrow. For $R : A \rightarrow B$ write $\exists : PB \rightarrow B$, dropping the subscript.

$$\in \triangleq \exists^\circ : A \rightarrow PA$$

(10b)

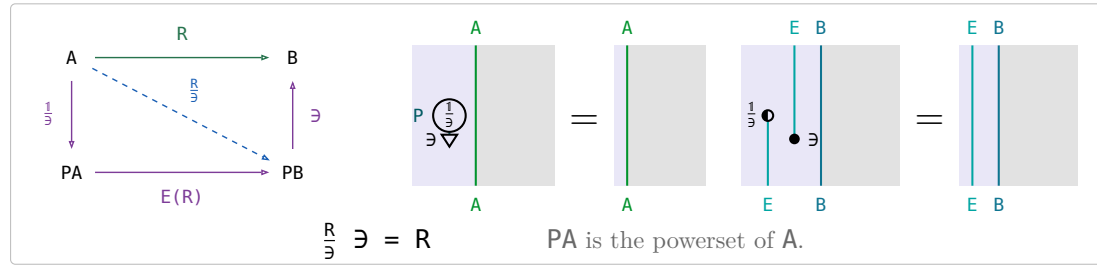
	law	the reading
1	$\exists_R \square = R\square, \exists_R = \exists_R \square$	One $\exists$ per object, not per arrow.
2	$\exists$ is thick	Comprehension: every x has a set of exactly the people x admires.
3	$\frac{R}{\exists} : A \rightarrow PB$, for $R : A \rightarrow B$	convert a relation to a function. $a \frac{R}{\exists} = \{b \mid a R b\}$
4	$\text{Map}(\frac{R}{\exists})$	
5	$\frac{R}{\exists} \exists = R$	reading the chosen set back through $\exists$ returns the relation.
6	$\text{Simple}(F) \implies F \sqsubseteq \frac{F \exists}{\exists}$	A partial choice of sets is inside the total one.
7	the singleton map is monic: $\frac{1}{\exists} (\frac{1}{\exists})^\circ \sqsubseteq \mathbb{1}$ two points with the same one-person set are the same point.	The one-person set.
8	$\frac{\exists}{\exists} = \mathbb{1}$	Make the set of a set, then read it back one level down.
9	fusion: $\text{Map}(f) \implies \frac{f R}{\exists} = f \frac{R}{\exists}$	Naturality of the unit, $\text{Map}(f) \implies f \frac{1}{\exists} = \frac{1}{\exists} E(f)$. renaming a point and then taking its one-person set is taking the set and renaming inside it.
10	$\text{Map}(f) \wedge \mathbb{1}\square = \exists\square \implies \frac{f}{\exists} = f \frac{1}{\exists}$	Rename first or take singletons first — the fusion row above at $R=1$.
11	$R\square = \exists\square \implies R /_s S = \frac{R}{\exists} (\frac{S}{\exists})^\circ$	x and y match when R sends x and S sends y to the same set.
12	$E(R) \triangleq \frac{\exists R}{\exists}$	$E(R) : PA \rightarrow PB$, image of a set of A
13	$\frac{R}{\exists} = \frac{1}{\exists} E(R)$	$\frac{1}{\exists} : x \mapsto \{x\}$
14	$\frac{S}{\exists} E(R) = \frac{S R}{\exists}$	absorption — the monad's composition law, $\frac{S}{\exists} \diamond \frac{R}{\exists} = \frac{SR}{\exists}$, §10.3
15	$E(R) = E(\frac{R}{\exists})$ union	the image of a set under R is the union of the R -sets of its points.
16	$\supseteq \triangleq \exists / \exists$	$xs \supseteq ys \iff \forall a. ys \exists a \rightarrow xs \exists a$
17	$\subseteq = \supseteq^\circ$	$xs \subseteq ys \iff ys \supseteq xs$
18	$T^\circ \frac{T}{\exists} \sqsubseteq \in$	every T -value of a point is a member of its T -set: $\wedge(T)$ is a map and $\wedge(T)\exists = T$.

(10c)



§10.1. $\mathbf{i}\text{-}\mathbf{E}$ Power Allegory defined as adjunction

(10.1a)



$\mathbf{i}$ is the inclusion $\mathbf{Map}(\mathcal{A}) \rightarrow \mathcal{A}$, doing nothing, so $\mathbf{E}$ is both the functor and the monad $\mathbf{iE}$.

§10.2. $\epsilon \setminus$ as a composite

$\epsilon \setminus Z = \subseteq (\frac{Z^\circ}{\exists})^\circ$ (10.2a)

$\mathbf{xS}$ is related to $\mathbf{c}$ by $\epsilon \setminus Z$ exactly when $\mathbf{xS} \subseteq \Lambda(Z^\circ)(\mathbf{c})$, the set of every $\mathbf{a}$ with $\mathbf{a} Z \mathbf{c}$

1	$\epsilon \setminus Z$	$Z = \epsilon \Lambda(Z^\circ)^\circ \xrightarrow{\frac{R}{\exists}} \exists = R$
2	$= \epsilon \setminus (\epsilon (\frac{Z^\circ}{\exists})^\circ)$	$X \setminus (Y f^\circ) = (X \setminus Y) f^\circ, f \text{ a map}$
3	$= \subseteq (\frac{Z^\circ}{\exists})^\circ$	

$R \setminus S = \frac{R^\circ}{\exists} \subseteq (\frac{S^\circ}{\exists})^\circ$ (10.2b)

$\mathbf{b}$ is related to $\mathbf{c}$ by $R \setminus S$ exactly when $\Lambda(R^\circ)(\mathbf{b}) \subseteq \Lambda(S^\circ)(\mathbf{c})$, the R -preimage of $\mathbf{b}$ inside the S -preimage of $\mathbf{c}$

1	$R \setminus S$	$R = \epsilon \Lambda(R^\circ)^\circ \xrightarrow{\frac{R}{\exists}} \exists = R$
2	$= (\epsilon (\frac{R^\circ}{\exists})^\circ) \setminus S$	$(XY) \setminus S = Y \setminus (X \setminus S)$
3	$= (\frac{R^\circ}{\exists})^\circ \setminus (\epsilon \setminus S)$	$f^\circ \setminus X = fX; \epsilon \setminus S = \subseteq \Lambda(S^\circ)^\circ \xrightarrow{\frac{Z^\circ}{\exists}} \epsilon \setminus Z = \subseteq (\frac{Z^\circ}{\exists})^\circ$
4	$= \frac{R^\circ}{\exists} \subseteq (\frac{S^\circ}{\exists})^\circ$	

§10.3. $\mathcal{A} \equiv \text{Kleisli}(E)$

(10.3a)

$$E(R) \triangleq \frac{\exists}{\exists} R, \quad \frac{1}{\exists} : A \rightarrow PA, \quad \text{union} = E(\exists) : PPA \rightarrow PA$$

$$f \diamond g \triangleq f \circ E(g) \circ \text{union}, \quad \text{for } f : A \rightarrow PB \text{ and } g : B \rightarrow PC$$

the monad is on $\text{Map}(\mathcal{A})$, not on the allegory: E is a relator on all relations, but $\frac{1}{\exists}$, union and $f \circ E(g) \circ \text{union}$ are maps, and the Kleisli construction happens where they live.

(10.3b)

$$\begin{aligned} \boxed{\frac{S}{\exists} \diamond \frac{R}{\exists}} & \xrightarrow{\text{definition of } \diamond, \text{ union}=E(\exists)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\right) E(\exists)} \xrightarrow{E \text{ a functor, } E(X)E(Y)=E(XY)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\exists\right)} \\ & \xrightarrow{\frac{R}{\exists} \exists = R, \frac{R}{\exists} \exists = R} \boxed{\frac{S}{\exists} E(R)} \xrightarrow{\frac{S}{\exists} E(R) = \frac{S}{\exists} R, \text{ absorption}} \boxed{\frac{SR}{\exists}} \end{aligned}$$

the first three steps only unfold $\diamond$ — E is a functor, so $E\left(\frac{R}{\exists}\right)\text{union}=E\left(\frac{R}{\exists}\exists\right)=E(R)$ and the union is gone.

Absorption, $\frac{S}{\exists} E(R) = \frac{S}{\exists} R$, is the whole law, and it is the functoriality of $\frac{\cdot}{\exists}$.

1 goes to $\frac{1}{\exists}$, the Kleisli identity, by definition — so $\frac{\cdot}{\exists}$ is an isomorphism of categories $\mathcal{A} \equiv \text{Kleisli}(E)$, and §10.1's $i \dashv E$ is the Kleisli adjunction. i is the identity on objects, so every object of the allegory is free.